

Aadarsha Gopala Reddy

+1 (740) 802-1776 | adurs2002@gmail.com | linkedin.com/in/agopalareddy | github.com/agopalareddy | agreddy.com

SUMMARY

Full-stack engineer with hands-on experience building Android apps, embedded C++ firmware, and production React interfaces for IoT device fleets — seeking full-stack/embedded software engineering roles.

SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, Java, Kotlin, Go, C++, C#, R.
Frameworks & Libraries: React, Node.js, FastAPI, Socket.IO, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Streamlit.
Tools & Technologies: Git, Docker, Linux, Arduino/Embedded Systems, L^AT_EX, AWS, PostgreSQL, MongoDB, Protobuf, Snowflake, Apache Airflow, Spark, Gemini/OpenAI APIs, Microsoft 365, Google Workspace.
Languages: Proficient in English, Kannada, Telugu; Conversational in Hindi.

EXPERIENCE

Washington University in St. Louis St. Louis, Missouri, USA
Graduate Teaching Assistant - CSE 5114: Data Manipulation and Management at Scale **January 2026 - May 2026**

- Taught large-scale data pipeline concepts (Snowflake, Airflow, Spark, Kafka, Flink) to a graduate cohort, covering fault-tolerant design, streaming windowing, and query optimization.

Master's Thesis Research **August 2025 - May 2026**

- Built an end-to-end pipeline processing 790,000 naturalistic driving trips from 304 older adults, engineering vehicle-agnostic weekly behavioral features and benchmarking six models under leave-one-participant-out cross-validation.

Crittero, Inc. Remote
AI Engineer Intern **June 2025 - August 2025**

- Generated 100,000 labeled social media posts across seven user personas via time-series behavioral modeling in TypeScript, capturing post sequences, engagement timings, and content preferences.
- Built a configurable Python CLI test harness (700+ lines) with deterministic seeding and CSV export for reproducible evaluation, validating a TensorFlow/Keras recommendation model (80% accuracy) across manually tuned configurations.

Lab714 Boca Raton, Florida, USA
Data Analyst Intern **June 2023 - May 2024**

- Designed and engineered the Figma-to-React UI for an all-in-one environmental sensor device, delivering a pixel-perfect, production-ready interface from initial wireframes.
- Built AWS data ingestion pipelines to normalize telemetry from a deployed IoT device fleet, powering real-time client dashboards that surfaced equipment alerts and cut pool maintenance costs by 20%.

PROJECTS

Personal Projects St. Louis, Missouri, USA

Blink Morse Decoder **June 2026**

- Built a real-time Android app (Kotlin, CameraX) that decodes deliberate eye blinks into Morse code via facial-landmark tracking at 30Hz and a debounced blink state machine filtering involuntary micro-blinks.

AURA: Ambient Wellness Companion **June 2026**

- Built a FastAPI backend orchestrating a Google ADK multi-agent system for elderly wellness check-ins, with least-privilege MCP toolsets and a deterministic privacy guard persisting only allowlisted patient telemetry.

EDUCATION

Washington University in St. Louis St. Louis, Missouri, USA
M.S. in Computer Science **August 2024 - May 2026**

- Thesis:* Toward Vehicle-Agnostic Driving Signatures for Cognitive Impairment Prediction from Naturalistic Driving Data

Ohio Wesleyan University Delaware, Ohio, USA
B.A. in Computer Science and Data Analytics **August 2020 - May 2023**

- Minor:* Economics | *Honors:* Golden Bishop Award, Mortar Board, Florence Leas Prize, Dean's List

LEADERSHIP

Washington University in St. Louis St. Louis, Missouri, USA
Vice President of the Graduate-Professional Council (GPC) Chamber, GPSC **May 2025 - May 2026**

- Chaired the Constitution Committee to draft and ratify founding governing documents; built committee rosters via ranked-choice preference allocation; represented graduate students to university leadership.